温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W04
Strategic Mgt Global Info Sys

Strategic Use of Information Resources

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Learning Objective

By the end of this session, students should be able to:

- Define what constitutes an **information resource**.
- Explain how information resources can be used strategically to attain and sustain competitive advantage.
- Apply key strategic frameworks – Porter's Five Forces, the Value Chain Model, and the Resource-Based View (RBV) – to analyze how IS enhances firm competitiveness.
- Evaluate the risks and challenges of depending on information systems for competitive advantage.

Information Resource

- Information resources, is defined as the available **data**, **technology**, **people**, and **processes** within an organization to be used by the manager to perform business processes and tasks.

Type	Definition	Examples
IT Assets	Tangible or intangible items the firm can use to create or deliver products.	IS infrastructure, databases, websites, equipment
IT Capabilities	Skills and know-how enabling the firm to use IT assets effectively.	Data analytics expertise, systems design, IT–business alignment

Information Resource – IT Assets

- IT Infrastructure
  - Foundation for the delivery of a firm's products or services.
  - *Examples:* hardware, software, data, network, website
- Information Repository
  - Logically related data captured, organized, and retrieved by the firm.
  - *Examples:* customer, employee marketplace, vendor information

Information Resource – IT Capability

- Technical Skills
  - Designing, developing, and implementing IS
  - *Examples:* systems analysis, programming, web design, data analysis, network design and implementation skills
- Management Skills
  - Managing the IT function and IT projects
  - *Examples:* business process knowledge, technology evaluation, project management, envisioning innovative IT solutions
- Relationship Skills
  - *Spanning* skills such as business-IT relationship management
  - *External* skills such as vendor and platform management

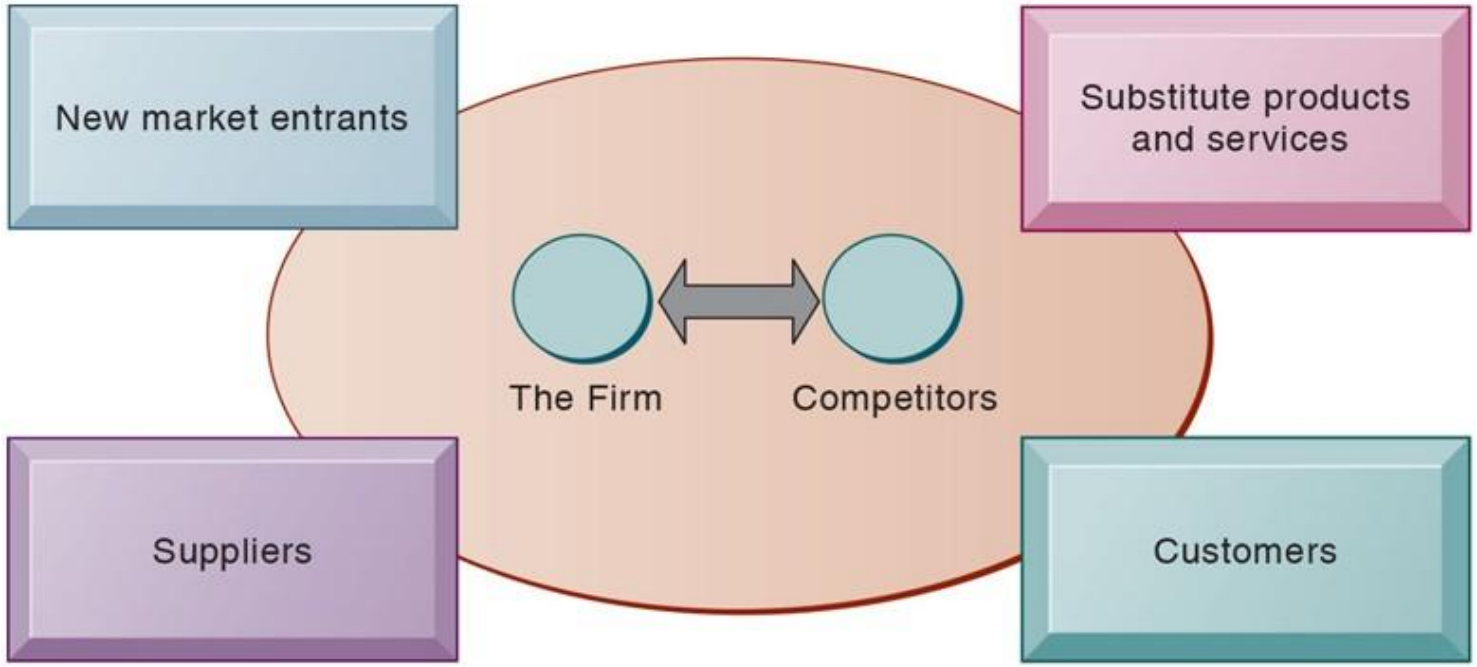
From Resources to Strategy

- In today's competitive environment, how should the information resources be organized and applied to enable the organization to compete most effectively?

- Managers must take multiple views of the strategic landscape, such as:
  - **First view:** *Porter's Five Forces* – analyze industry competition
  - **Second view:** *Porter's Value Chain* – identify where IS add value
  - **Third view:** *Recourse-Based View (RBV)* – assess strategic IS resources and capabilities

Michael Porter's Competitive Forces Model

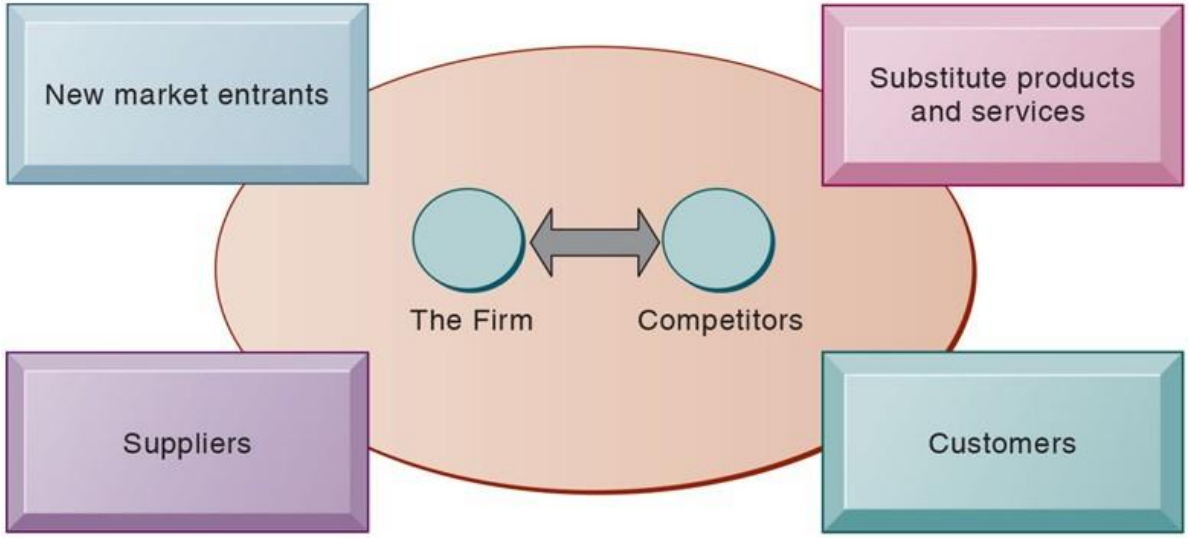
1. Why do some firms perform better than others?
2. How can organizations analyze and build such advantages?



What is it? Why does it matter? How do we use it?

Michael Porter's Competitive Forces Model

- What is it?
  - A framework for analyzing the competitive environment of a business.
  - Help assess how five key forces determine an industry's profitability and attractiveness.
- Why it matters?
  - Helps firms understand where competitive pressures come from.
  - Identifies strategic levers – areas to defend, differentiate, or invest – to carve out advantage.
  - Adapt business and information systems strategies in light of industry dynamics.



Michael Porter's Competitive Forces Model

Force 1: Competitive Rivalry

- What it is
  - Measures the intensity of competition among existing firms in an industry.
  - Includes price wars, advertising battles, service or product innovations, and frequent strategic moves.
- Why it matters
  - Usually the **strongest** force shaping industry profitability.
  - High rivalry → lower margins and reduced returns.
  - Success depends on understanding competitors' positions, pricing, and marketing strategies.
- Key drivers of rivalry
  - Innovation and speed of product improvement
  - Online vs. offline competition
  - Advertising intensity and brand visibility
  - Cost leadership vs. differentiation strategy
  - Industry concentration ratio (number and size of competitors)

Michael Porter's Competitive Forces Model

Force 2: Threat of New Entrants

- What it is
  - Refers to the ease or difficulty with which new competitors can enter an industry and challenge existing firms.
  - New entrants bring new capacity, resources, and ideas, which can pressure prices and profitability.
- Why it matters
  - High entry threat → limits pricing power and increases competition.
  - Low entry threat → allows established firms to maintain stable margins and defend market share.
- Key barriers to entry
  - Economies of scale and cost advantages
  - Brand loyalty and customer switching costs
  - Capital investment requirements
  - Access to distribution channels and supplier relationships
  - Government regulation or licensing

Michael Porter's Competitive Forces Model

Force 3: Threat of Substitutes

- What it is
  - The extent to which alternative products or services can satisfy the same customer need.
  - Substitutes may come from outside the industry and change customer preferences or price sensitivity.
- Why it matters
  - High threat of substitutes → limits pricing power and reduces industry profitability.
  - Firms must differentiate, innovate, or add value to retain customers.
- Key drivers of substitution threat
  - Availability and affordability of alternatives
  - Switching costs for customers
  - Performance or convenience of substitutes
  - Technological change enabling new options
  - Customer loyalty and perceived uniqueness

Michael Porter's Competitive Forces Model

Force 4: Bargaining Power of Buyers

- What it is
  - Describes the ability of customers to influence prices, quality, and terms.
  - Powerful buyers can demand lower prices, better service, or higher product quality.
- Why it matters
  - High buyer power → reduces industry profitability.
  - Firms must strengthen customer relationships or differentiate offerings to reduce dependence.
- Key drivers of buyer power
  - Buyer concentration vs. supplier concentration (few large buyers = more power)
  - Product standardization and low switching costs
  - Price sensitivity of buyers
  - Availability of substitutes or alternative suppliers
  - Importance of product quality or brand to buyer's success

Michael Porter’s Competitive Forces Model

Force 5: Bargaining Power of Suppliers

- What it is
  - Refers to the ability of suppliers to influence prices, quality, and availability of inputs.
  - Strong suppliers can demand higher prices or limit access to key resources.
- Why it matters
  - High supplier power → increases costs and reduces profitability for firms.
  - Managing supplier relationships is critical for cost control and supply stability.
- Key drivers of supplier power
  - Supplier concentration vs. industry demand
  - Uniqueness or scarcity of inputs
  - Switching costs for firms to change suppliers
  - Availability of substitutes for inputs
  - Supplier integration (forward integration into the buyer’s market)

Leveraging IS for Competitive Advantage

- Key Question
  - What can a firm do when facing intense competition, substitutes, and new entrants?
- Strategic Answer
  - Leverage Information Systems to enable one or more of Porter's Four Generic Strategies:
    - Low-Cost Leadership
    - Product Differentiation
    - Focus on Market Niche
    - Customer & Supplier Intimacy
- Key Idea
  - Information systems are not just support tools, they are **strategic enablers** for competitive advantage.

Leveraging IS for Competitive Advantage

IS Strategy 1 — Low-Cost Leadership

- Goal
  - Use information systems to achieve lowest operational costs and offer lowest prices in the industry.
- Example: Walmart
  - Built a real-time inventory and replenishment system linking checkout data directly to suppliers.
  - Point-of-sale (POS) terminals send sales data instantly to headquarters.
  - Suppliers access live inventory and sales data to restock efficiently.
  - Enables just-in-time replenishment, reducing warehouse costs and stockouts.
  - Results in lower overhead, better demand alignment, and sustained price leadership.
- Key Concept

  → Walmart's system is an Efficient Customer Response (ECR) model — directly linking customer purchases to production and supply chain operations.

Leveraging IS for Competitive Advantage

IS Strategy 2 — Product Differentiation

- Goal
  - Information systems help firms create, enhance, or personalize products and services to stand out from competitors.
- Example of IS-driven differentiation
  - Tech leaders (Google, Meta, Amazon, Apple)
  - → Invest billions in continuous product innovation and service enhancement.
  - → Example: Apple's yearly iPhone and iPad upgrades.
  - Crayola→ Expands from crayons to digital creative tools for children.
  - PepsiCo – Doritos Silent→ Uses AI to improve the customer experience rather than the product itself.
- Takeaway:

  ☞ Differentiation now extends beyond product features — it's about the entire customer experience, enabled by information systems.

Leveraging IS for Competitive Advantage

IS Strategy 3 — Focus on Market Niche

- Goal
  - Information systems and analytics enable firms to identify, understand, and serve specific customer segments better than competitors.
- How IS supports niche focus
  - Collect and analyze data from multiple sources
    → credit card transactions, online behavior, demographics, retail purchases
  - Use AI / ML to detect patterns and predict preferences
  - Enable one-to-one marketing and personalized offers
  - Power AI-enhanced CRM systems for customer profiling and engagement
- Examples

  🏀 Charlotte Hornets (NBA) – combine ticket, merch, and food purchase data → personalized fan profiles and targeted promotions

  💳 Credit card companies – use AI/ML to identify profitable or risky cardholders

Leveraging IS for Competitive Advantage

IS Strategy 4 — Strengthen Customer and Supplier Intimacy

- Goal
  - Information systems enhance collaboration, coordination, and personalization, creating mutual value and long-term loyalty.
- Supplier Intimacy
  - Share real-time production schedules and inventory data
  - Enable suppliers to plan and ship proactively
  - Example: 🚗 Toyota & Ford – suppliers access systems directly to synchronize production
- Customer Intimacy
  - Track preferences, purchase history, and behavior
  - Provide personalized recommendations and services
  - Example: 🛒 Amazon – uses data to recommend items and anticipate needs

Value Chain Model

- Porter's Competitive Forces Model
  - Analyzes external competition and identify generic strategies (low-cost leadership, product differentiation, focus market niches, strengthen customer and supplier intimacy).
  - However, it does not specify how to operationalize those strategies inside the firm.

- If your goal is to achieve operational excellence, where do you start?

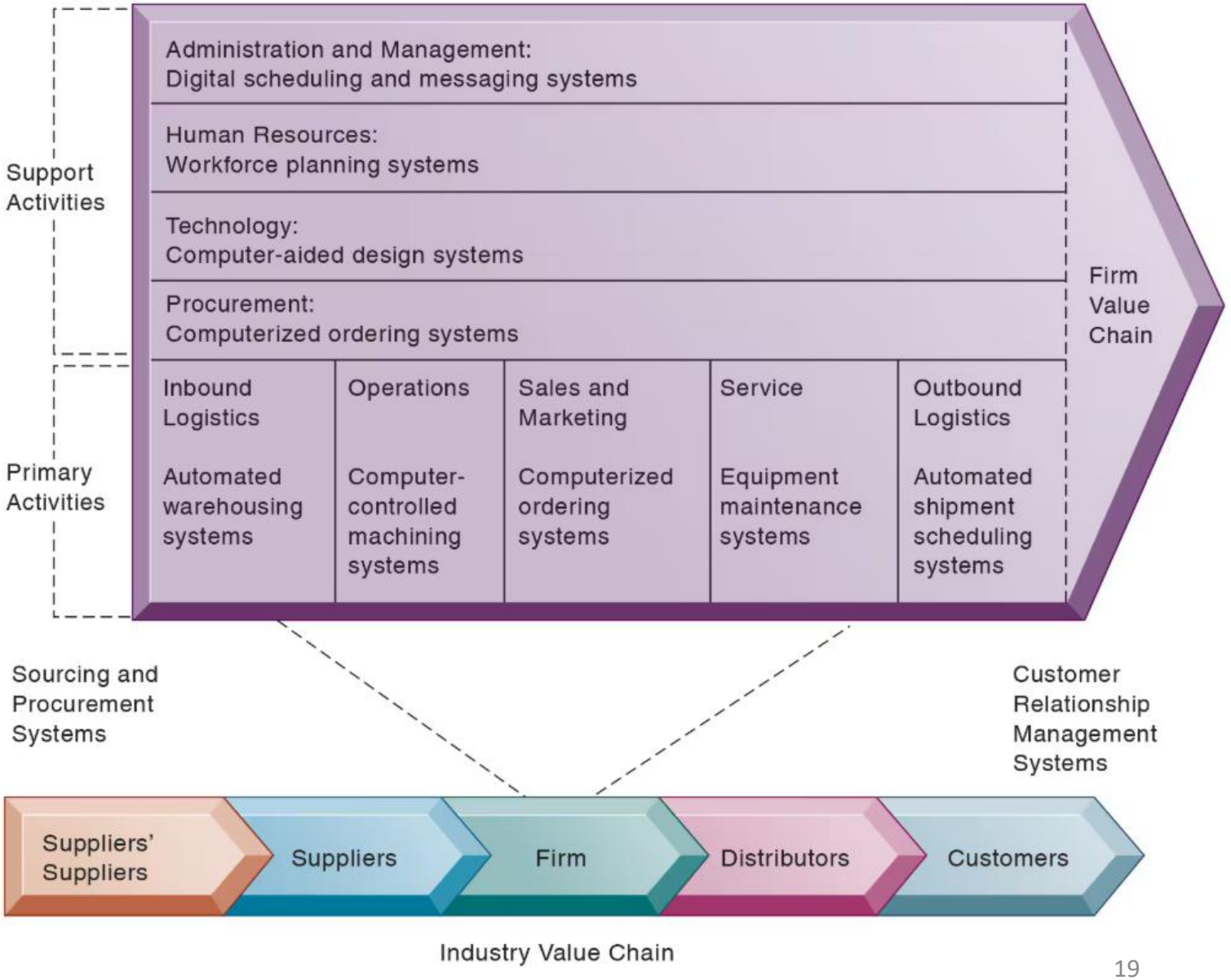
→ The Value Chain Model (Porter, 1985)
  - Provides a roadmap for operational excellence by examining a series or chain of basic activities that add a margin of value to a firm's products and services.
  - Identifies leverage points where information systems can have the greatest strategic impact.

Value Chain Model

Primary activities directly create customer value through core processes:

→ Inbound logistics, operations, outbound logistics, sales & marketing, and service.

(From receiving materials → producing goods → delivering → selling → maintaining products.)

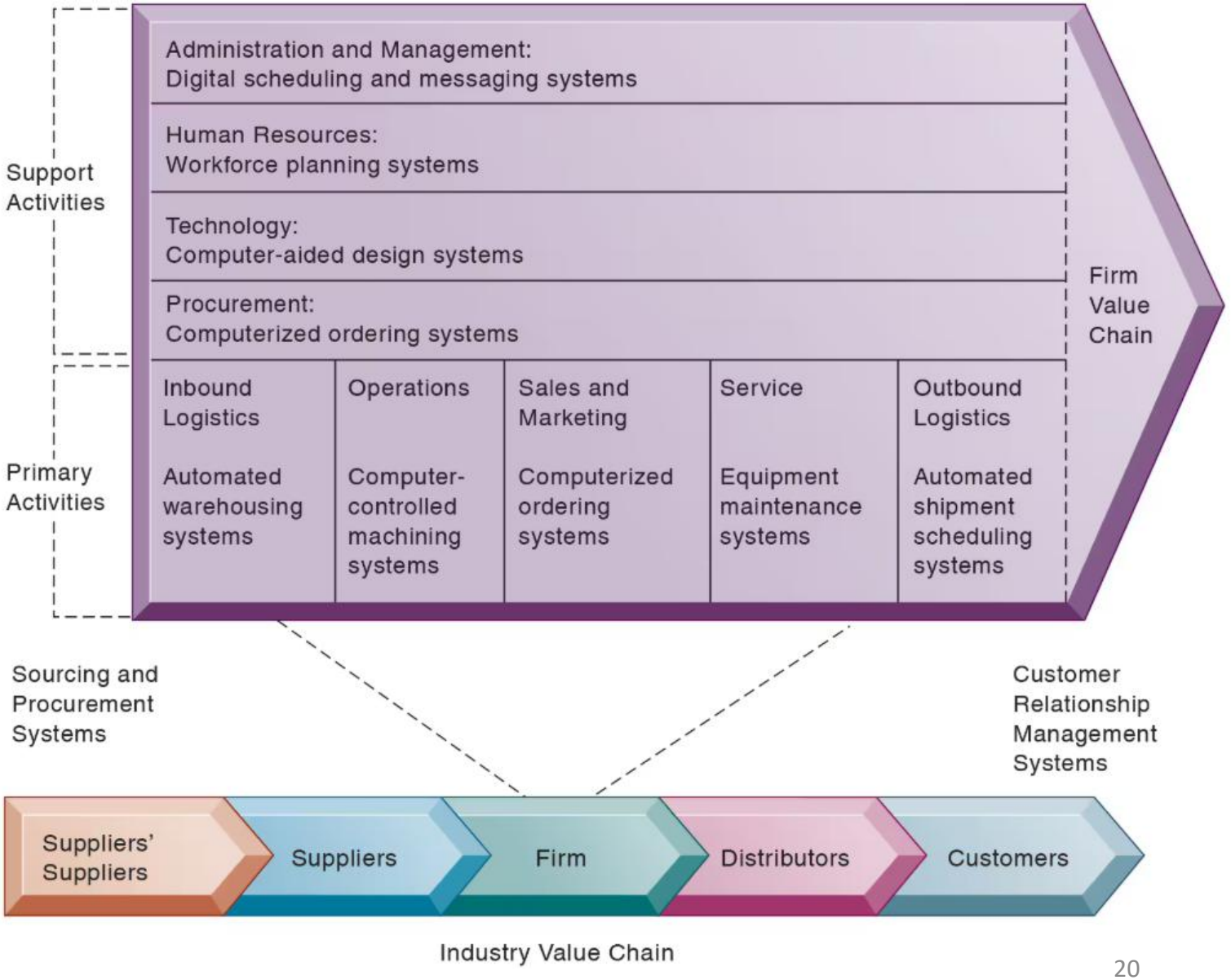


Value Chain Model

Support activities enable and enhance primary activities through:

→ Infrastructure, human resources, technology development, and procurement.

(They provide the foundation that allows core operations to run efficiently.)



Value Chain Model – Zara

Primary Activity	Zara's Value Chain
Inbound Logistics	• IT-enabled JIT (just in time) strategy • Most dyes are purchased from Zara subsidiaries • Suppliers are located near production facilities
Operations	• IT supports decisions about fabric, cut, and pricing • Products are hung on hangers to avoid ironing at stores • Price tags are already on the products • 50% of merchandise is produced in house • Fabric is cut and dyed by robots in 12 highly automated factories
Outbound Logistics	• Clothes move on miles of conveyer belts • Clothes reach stores within 48 hours of the order
Marketing and Sales	• Limited inventory permits low percent of unsold goods • Point-of-Sales systems linked to HQ • Managers provide information on customer needs daily via hand-held computers
Service	• No focus on service; clothes only last 10 wears

Value Chain Model – Zara

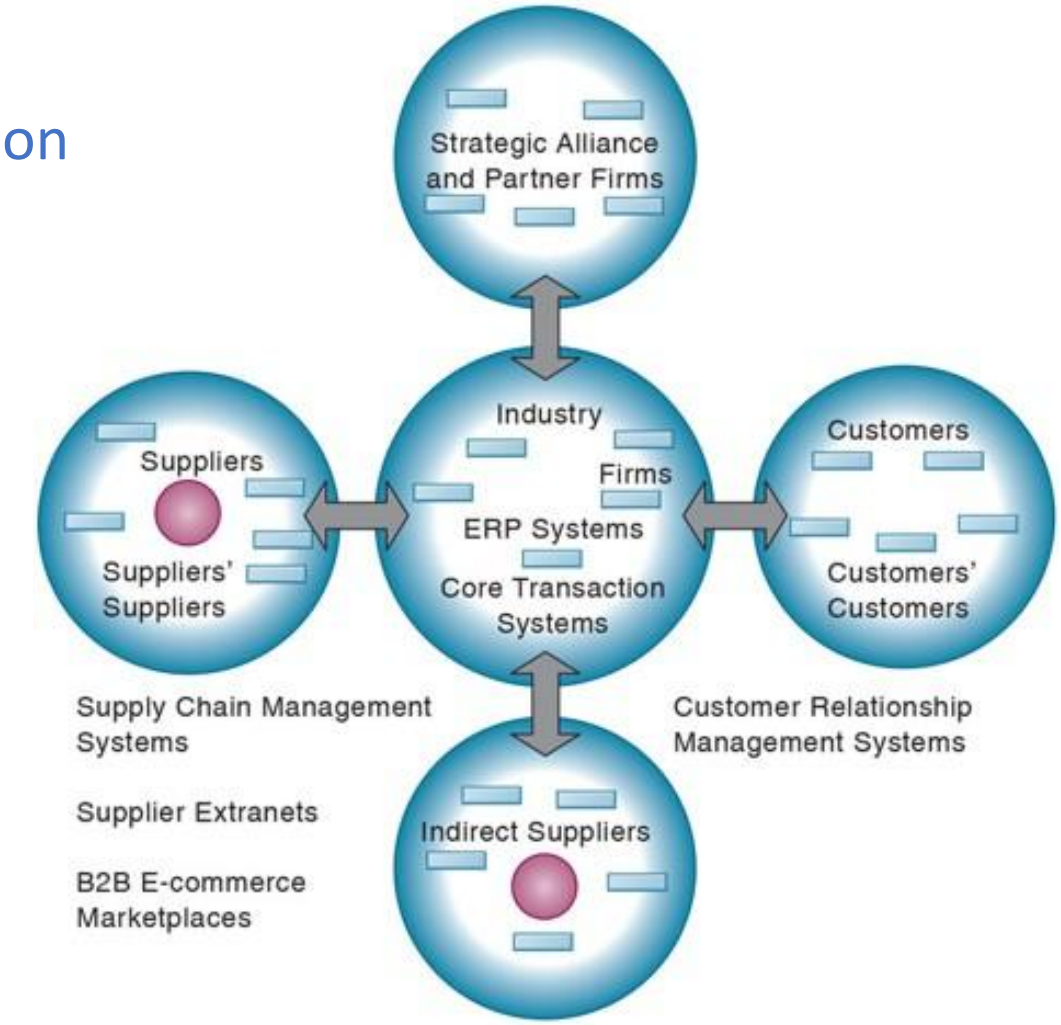
Secondary Activity	Zara's Value Chain
Organization	• Tightly-knit collaboration between designers, store managers, market specialists, production managers, production planners
Human Resources	• Managers are trained to understand customer needs and report to designers daily • Managers help Zara keep on cutting edge of fashion
Technology	• All primary activities are supported by integrated technology • Automated conveyers speed up logistics
Purchasing	• Vertical integration reduces the amount of purchasing needed

The Value Web

From internal efficiency to inter-firm coordination

The Value Web

- A network of independent firms using IT to coordinate their value chains to produce a product or service for a market collectively.
- Non-linear, customer-driven, flexible, and adaptive to market changes.
- Enables faster response, time-to-market, and collaborative innovation.



From Attaining to Sustaining Competitive Advantage

- Key Question
  - How can firms ***maintain*** the advantages they've worked hard to build – especially in fast-changing, tech-driven industries?
- The Challenge of Sustainability
  - In many industries, advantages erode quickly due to hypercompetition.
  - Example: Banking evolution
    - → 1960s: in-person banking only
    - → 1970s: ATMs
    - → 1980s: phone banking
    - → 1990s: web banking
    - → 2000s–present: mobile banking
  - Each innovation was **quickly imitated**, shortening competitive lead time.

Can IT/IS help?

- The Cost of Keeping Up
  - Some technologies **reduce costs** (e.g., ATMs replacing tellers).
  - Others **raise costs** by forcing multi-channel investment (phone, web, mobile).
  - Small banks face rising **technology burden** → fewer can compete.

A deadly distraction?

Building Barriers to Sustain Advantage

Barrier	Definition	Examples
IT Project Barrier	It would be a large undertaking for a competitor to build the system to copy the capability.	• Requires a large investment • Requires a long time to build • Complicated to build
IT Assets and Capabilities Barrier	Competitors might lack the IT resources to copy the capability.	• Database of customers that cannot be copied • Expert developers or project managers
Complementary Resources Barrier	The firm has other resources that create a synergy with the IT that provides competitive advantage.	• Respected brand • Partnership agreements • Exclusivity arrangements • Good location
Preemption Barrier	The firm “got there first.”	• Loyal customer base built at the beginning • Firm known as “the” source

Piccoli, G., & Ives, B. (2005). IT-dependent strategic initiatives and sustained competitive advantage: A review and synthesis of the literature. MIS Quarterly, 747-776.

Innovation to Sustain Advantage

- Even the strongest barriers erode over time. Firms must keep reinventing products, services, and even their business models.
- How innovation sustains advantage
  - Use IT to improve offerings and enhance customer experience.
  - Anticipate disruption — don't just respond to it.
  - Balance short-term improvements with long-term transformation.
- Example: Netflix
  - Moved from DVD delivery → Streaming → Original content & film production.
  - *Embraced revenue decline* to lead the next wave.
  - Turned short-term risk into industry-wide transformation.
- Takeaway

  🔁 Sustainability ≠ stability — it means continuous renewal through innovation.

Resource-Based View (RBV)

- Purpose
  - Evaluate whether a firm's **information resources** attain and sustain competitive advantage.
- Core Idea
  - Focuses on resources, not just activities or market position.
  - A resource must be **valuable**, **rare**, **inimitable**, and **non-substitutable** (VRIN) to yield sustained advantage.
  - IS resources include *data, systems, processes, culture, skills, and relationships*.

Resource-Based View (RBV)

To **Attain** Advantage	To **Sustain** Advantage
Valuable & Rare → improve efficiency, effectiveness, innovation	Hard to Imitate, Substitute, or Transfer → embedded in processes, relationships, culture
Example: Early mobile banking apps created value but soon became “table stakes.”	*Example:* Walmart’s logistics IS or firm-specific analytics routines are hard to copy.

Resource-Based View (RBV) - Zara

	Value Creation		Value Sustainability		
Resource/Attribute	**Value**	**Rarity**	**Imitation**	**Substitution**	**Transfer**
IT Asset					
IT Infrastructure	**Moderate** because of its skillful use of the POS equipment, handheld computers, automated conveyors, and computer-controlled equipment to cut patterns, but similar technology could be purchased and used by competitors		**Easy** to imitate and transfer its infrastructure **Moderate** for substitution of infrastructure (automated conveyers)		
Information Repository	**High** value and rarity because of its information about customers' preferences and body types, which Zara leverages strategically; well integrated with Zara's operations and personnel; retail information analyzed by designers to identify future products		**Difficult** to imitate and transfer **Extremely difficult** to substitute because of the volume and nature of the data		

Resource-Based View (RBV) - Zara

	Value Creation		Value Sustainability		
Resource/Attribute	**Value**	**Rarity**	**Imitation**	**Substitution**	**Transfer**
IT Capability					
Technical Skills	**Low** value/rarity because IS professionals could be hired relatively easily to perform the technical work		**Moderately difficult** to imitate, substitute, or transfer; some sustainability results because the skills are used to integrate across a range of systems		
IT Management Skills	**High** value/rarity because they were acquired over time		**Difficult** to imitate, substitute, or transfer; resources leveraged well		

Resource-Based View (RBV) - Zara

	Value Creation		Value Sustainability		
Resource/Attribute	**Value**	**Rarity**	**Imitation**	**Substitution**	**Transfer**
IT Capability					
Relationship Skills — *Externally Focused*	**High** value from relationships with European manufacturers **Moderate** rarity because other companies also have relationships with manufacturers although required time to develop the relationship		**Difficult** to imitate, substitute, or transfer; turnaround time of under 5 weeks from conception to distribution		
Relationship Skills — *Spanning*	**High** rarity of spanning		**Difficult** to imitate, substitute, or transfer spanning; unusual tight-knit teams at headquarters not easy to imitate or purchase in the marketplace, allowing the ability to correctly interpret and quickly respond to customer needs		

Potential Risk

- There are many potential risks that a firm faces when attempting to use IT to outpace their competition.
- Executives should be aware of these risks in advance!
  - Awakening a sleeping giant – a large competitor with deeper pockets may be nudged into implementing IS with even better features (Schwab/eTrade; FedX/UPS)
  - Demonstrating bad timing – sometimes customers are not ready to use the technology designed to gain strategic advantage (Apple's Newton, Shelternet)
  - Implementing IS poorly – information systems can fail because they are poorly implemented (Nike; Hershey)
  - Failing to deliver what users want – systems that don't meet the firm's target market likely to fail (Streamline.com groceries)
  - Running afoul of the law – Using IS strategically may promote litigation (Airline reservation systems)

Summary

- Information resources include data, technology, people, and processes — they can be assets or capabilities.
- When aligned with business strategy, IS can create and sustain competitive advantage.
- Managers analyze information resource strategically through three key frameworks:
  - **Porter's Five Forces** – understand industry structure and competitive pressures
  - **Porter's Value Chain** – identify where IS adds value across activities
  - **Resource-Based View (RBV)** – assess valuable, rare, and hard-to-imitate resources
- Sustaining advantage requires continuous innovation and effective resource management.
- Managers must also recognize and mitigate risks associated with IS-driven strategies.

Daily Feedback